Spring Boot Complete Guide

Version 1.0 - PrepAI Knowledge Base

1. Introduction

Spring Boot is a Java framework that simplifies building production-ready applications with minimal configuration.

2. Spring vs Spring Boot

Spring requires more configuration, while Spring Boot provides auto-configuration, starter dependencies, and embedded servers.

3. Core Concepts

Dependency Injection (DI), Inversion of Control (IoC), Beans, Application Context.

4. Project Structure

Typical Maven/Gradle project with src/main/java, src/main/resources, application.properties and pom.xml.

5. Starter Dependencies

spring-boot-starter-web, spring-boot-starter-data-jpa, spring-boot-starter-security, spring-boot-starter-test.

6. REST API Development

Use @RestController, @RequestMapping, @GetMapping, @PostMapping, @PutMapping and @DeleteMapping.

7. Spring Data JPA

Repositories, Entity classes, CRUD operations, Query Methods.

8. Exception Handling

Use @ControllerAdvice and @ExceptionHandler for centralized exception handling.

9. Configuration

application.properties / application.yml, Profiles, Environment Variables.

10. Spring Boot Interview Questions

- What is Spring Boot?
- Difference between Spring and Spring Boot?
- What is Dependency Injection?
- What is @Autowired?
- Explain @RestController.
- What is Spring Data JPA?
- Difference between @Component, @Service and @Repository?
- What is application.properties?
- What is Actuator?

• What is Spring Boot Starter?

11. Best Practices

Use layered architecture, constructor injection, DTOs, validation, logging and global exception handling.

12. Mini Project Idea

Build a Student Management REST API using Spring Boot, MySQL and JPA with CRUD operations.